Csaba Purszki

SENIOR C++ SYSTEMS ENGINEER

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SUMMARY

Senior C++ systems engineer with over 20 years of experience in real-time, low-latency, and safety-critical software. My main technical areas are modern C++, concurrency, lock-free design, memory management, shared-memory IPC, and cross-platform architecture. I have designed reusable frameworks and system components for automotive ADAS products used in mass production. I am now applying this systems-level C++ expertise to Bitcoin protocol research and to new challenges in financial technology and distributed systems.

CORE COMPETENCIES

Modern C++ — STL, RAII, templates, low-level and cache-aware optimization.

Low-Latency & Lock-Free — atomics, memory ordering, lock-free data structures, predictable real-time hot paths.

Concurrency Verification — instrumented, reproducible tests for race and memory-order correctness, supported by static analysis.

Performance Engineering — profiling, custom allocators, deterministic O(1) allocation, memory efficiency.

Architecture & Systems — binary compatibility, semantic versioning, IPC / shared memory, cross-platform builds.

Technical Leadership — architecture reviews, mentoring, TDD, documentation, reusable engineering practices.

SELECTED ACHIEVEMENTS

- **Custom Memory Allocator** — Designed a custom allocator that avoids fragmentation and provides O(1) allocation, delivering deterministic performance in a mission-critical real-time system.
- **Deterministic Shared-Memory IPC** — Designed a low-latency, single-writer/multi-reader shared-memory queue for real-time ADAS surround view, covering lifecycle, data integrity, timing, and timestamped history; design parallels concepts later adopted by iceoryx2.
- **Patented Parking-Slot Detection** — Designed the software architecture of a camera-based parking-slot-marking detection algorithm used in mass-production ADAS; the work contributed to a patent.
- **C++ protobuf layer for nanopb** — Added a C++ layer to nanopb to make protocol-buffer integration easier in embedded projects.
- **Light-Client Block Filter Research** — Compared BIP 158 and Binary-Fuse filters across 50,000 mainnet blocks and 10 wallet scenarios. Fuse16 kept zero false negatives while cutting client-side query CPU by 9–81× on desktop and 6–46× on a Raspberry Pi 5, at bandwidth similar to GCS.

EXPERIENCE

Valeo Vision Systems — Independent Contractor (Computer Vision Kft) **05/2014 – Present**

Budapest, Hungary · Ireland — Automotive / Computer Vision

Algorithm Framework Expert

- Developed a reusable algorithm framework used by multiple teams for memory pools, debugging and visualization, logging, and shared-memory I/O.
- Provided technical leadership, mentored junior engineers, and introduced good practices for concurrency and resource management.
- Improved team efficiency by introducing test-driven development with GoogleTest.

Algorithm Embedded Lead — Parking Slot Marking Detection

- Designed the software architecture of the patented Parking Slot Marking Detection algorithm.
- Helped develop a package-manager-based build system using semantic versioning and CI/CD for stable integration of multiple modules.
- Made backward compatibility easier, reduced integration work, and simplified releases.

Valeo Vision Systems — Software Engineer **05/2011 – 05/2014**

Tuam, Co. Galway, Ireland — Automotive / Computer Vision

DSP Lead — Calibration and Scene Viewing Group

- Designed the software architecture of computer vision algorithms.
- Helped define development processes and code-reuse practices.

Algorithm Integrator — Software Group

- Integrated and maintained an automatic road-based camera calibration algorithm across customer projects.
- Refactored and ported the algorithm from C to C++ as a reusable library optimized for TI VisionMid and OMAP platforms.

Intellio Zrt — Algorithm Developer **08/2005 – 04/2011**

Budapest, Hungary — Video Surveillance Systems

- Designed an algorithm development framework for object-tracker-based applications.
- Developed computer vision algorithms: object tracker, background subtraction, camera calibration, color recognition, shadow filtering.
- Developed a traffic detector that used statistical traffic-flow analysis to identify stopped vehicles and vehicles moving in the wrong direction.

PATENTS

Contributor to a German patent related to parking-slot detection (DE102019102561A1, “Verfahren zur Erkennung einer Fahrbahnmarkierung”).

EDUCATION · LANGUAGES · OTHER

Education & Dev

- Chaincode Labs BOSS — selected participant, 2026
- MSc in Electrical Engineering, BME, 2000–2005

Languages

- Hungarian (native)
- English (fluent)
- German (basic)

Other

- Traditional musician
- Weekend sailor